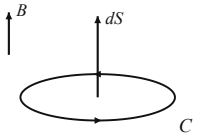
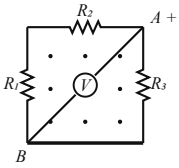
1 · Signs and directions



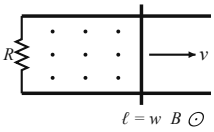
- Fingers along C, thumb $= d\mathbf{S}$. $\mathcal{E}$, $I > 0$: **along** C.
- **Lenz:** I fights the **change** of Ψ: a check, not a second minus.
- Wire: thumb along I, fingers $= \mathbf{B}$. Loop: fingers along I, thumb $= \mathbf{B}$ inside.
- Sheet: $\mathbf{H} = \frac{1}{2}\mathbf{J}_s \times \hat{n}$, $\hat{n}$ **from the sheet to the point**; $\mathbf{H}^+ - \mathbf{H}^- = \mathbf{J}_s \times \hat{n}_{-\to+}$.
- Parallel currents attract. Sheet force: $\mathbf{J}_s \times \mathbf{B}_{\text{others}}$ (average of both sides).
- $\mathbf{E} \times \mathbf{H}$ along the travel; $f(t - z/v)$ and $\omega t - \beta z$ go to $+z$.

2 · Faraday in 5 steps, and voltmeters



- (1) pick C; RHR: $d\mathbf{S}$ (2) $\Psi = \int \mathbf{B} \cdot d\mathbf{S}$ where $B \neq 0$ (3) $\mathcal{E} = -N\dot{\Psi}$ (4) $I = \mathcal{E}/R_{\text{tot}}$ (5) Lenz.
- **Voltmeter** reads $\int_+^- \mathbf{E} \cdot d\mathbf{l}$ **along its leads**. Leads + part of the circuit make a loop: $V + \sum(\pm IR) = -\dot{\Psi}_{\text{encl}}$ ($d\mathbf{S}$ by RHR on that loop); $+IR$ walking with I.
- Hugging a resistor: $\pm IR$. Square's diagonal: half the flux. Two routes around the flux differ by $\mathcal{E}$. Symmetric ring, diameter: 0.

3 · Motional emf

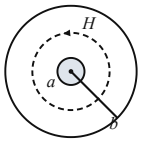


- $\mathcal{E} = \oint(\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$ along C. Bar: $vB\ell$, B **at the bar**; $\mathbf{v} \times \mathbf{B}$ pushes $+q$.
- Rails: $I = vB\ell/R$, $F = I\ell B$ against $\mathbf{v}$, $P = Fv = I^2R$.
- Square in $B_0(1 + x/L)\hat{z}$ at $v\hat{x}$: $\mathcal{E} = -s^2B_0v/L$, constant; moving along y: 0.
- Rod $\frac{1}{2}B\omega L^2$. Solenoid (a, $\dot{B}$): $E_\phi = -\frac{r}{2}\dot{B}$ in, $-\frac{a^2}{2r}\dot{B}$ out.

4 · Fields you must know

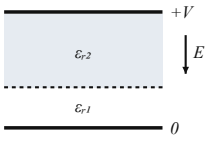
- H: solenoid nI; toroid core $NI/2\pi r$; coax $I/2\pi r$; thick wire $Ir/2\pi a^2$; zero outside the first three.
- $d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,d\mathbf{l} \times \hat{R}}{R^2}$, $d\mathbf{F} = I\,d\mathbf{l} \times \mathbf{B}$. Loop axis $\frac{\mu_0 I a^2}{2(a^2+z^2)^{3/2}}$, center $\frac{\mu_0 I}{2a}$. Wires $\frac{\mu_0 I_1 I_2}{2\pi d}$ N/m.
- Sheets $\pm J_s$: $H = J_s$ between, 0 out. Slab: a sheet JW outside, $dH_y/dz = -J_x$ inside. Net K: far field $\pm K/2$.
- $\delta(z)$ is 1/m: $K\delta(z-a)$ = sheet, $I\delta(x)\delta(y)$ = wire, $u(z-a) - u(z-b)$ = slab. $\frac{d}{dz}|z| = \operatorname{sgn} z$.

5 · Inductance



- $I \to \mathbf{H}$ (free I) $\to \mathbf{B} = \mu\mathbf{H} \to \Psi$ of one turn $\to L = N\Psi/I$.
- Solenoid $\frac{\mu N^2 A}{\ell}$; toroid $\frac{\mu N^2 h}{2\pi}\ln\frac{b}{a}$; coax $L' = \frac{\mu}{2\pi}\ln\frac{b}{a}$; pp $L' = \frac{\mu d}{w}$; layers: $\sum \frac{\mu_i}{2\pi}\ln\frac{r_{i+1}}{r_i}$.
- $L'C' = \mu\epsilon$. $M = \Psi_{21}/I_1 = M_{12}$; one turn on the core: L/N. $\frac{1}{2}LI^2$.
- L never depends on I; $\propto N^2$; all lengths $\times k$: $L \times k$, L' same.

6 · C, G, relaxation



- pp $\frac{\epsilon A}{d}$; coax $C' = \frac{2\pi\epsilon}{\ln(b/a)}$; spheres $\frac{4\pi\epsilon ab}{b-a}$, alone $4\pi\epsilon a$; $R = \frac{\ell}{\sigma A}$.
- $G/C = \sigma/\epsilon$, $\tau = RC = \epsilon/\sigma$. In DC a lossy C is just $1/G$.
- Stacked: same D, $V = D\sum\frac{d_i}{\epsilon_i}$. Side by side: same E, C's add.
- Leaky, steady: $\sigma_1E_1 = \sigma_2E_2$, $\rho_s = D_{2n} - D_{1n}$; insulator + leaky: all V on the insulator.
- $\rho = \rho_0 e^{-t/\tau}$, then integrate $\partial_z E_z = \rho/\epsilon$. Conductor: $\mathbf{E} = 0$, $\rho = 0$ inside.

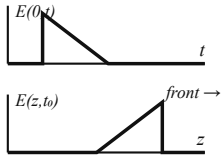
7 · Boundaries and materials

- E_t, B_n continuous; D_n jumps ρ_s, H_t jumps J_s. With no ρ_s, J_s: tangential D, B and normal E, H scale by the ϵ or μ ratio. $\frac{\tan\theta_2}{\tan\theta_1} = \frac{\mu_2}{\mu_1}$.
- $\mathbf{H}$: free currents only. $\mathbf{M} = (\mu_r - 1)\mathbf{H}$, $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$, bound sheet $\mathbf{M} \times \hat{n}_{\text{out}}$; $\mathbf{P} = (\epsilon - \epsilon_0)\mathbf{E}$. Dia $\chi_m < 0$, para small > 0, ferro $\gg 1$.

8 · Displacement current, curls

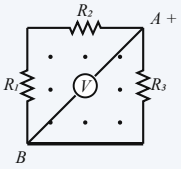
- $\mathbf{J}_d = \partial\mathbf{D}/\partial t$; gap current $C\dot{V}$ = wire current. Round plates: $H_\phi = \frac{r}{2}\epsilon\dot{E}$ in, $\frac{R^2}{2r}\epsilon\dot{E}$ out. $J_c/J_d = \sigma/\omega\epsilon$.
- $\nabla \times [f(w)\hat{e}] = f'(w)\,\hat{w} \times \hat{e}$**.** Curl amplitude $\beta\times$ amplitude, then $\div\mu\omega$ or $\epsilon\omega$: $E_0 = \eta H_0$, $\hat{\mathbf{H}} = \hat{\mathbf{k}} \times \hat{\mathbf{E}}$.
- **Cylindrical** (tested): $\nabla \times (H_\phi\hat{\phi}) = \frac{\hat{z}}{r}\frac{d(rH_\phi)}{dr}$, $\nabla \times (H_z\hat{z}) = -\hat{\phi}\frac{dH_z}{dr}$, $\nabla \cdot (E_r\hat{r}) = \frac{1}{r}\frac{d(rE_r)}{dr}$. Check $\nabla \cdot \mathbf{B} = 0$; $\mathbf{J} = \nabla \times \mathbf{H}$. Vector potential: not tested.

9 · Waves: read and sketch



- $v = \frac{\omega}{\beta} = f\lambda = \frac{c}{\sqrt{\mu_r\epsilon_r}}$, $\eta = \eta_0\sqrt{\frac{\mu_r}{\epsilon_r}}$, $\lambda = \frac{2\pi}{\beta}$.
- Sheet: $\mathbf{E} = -\frac{\eta}{2}\mathbf{J}_s(t - \frac{|z|}{v})$ **both sides**; $|H| = \frac{J_s}{2}$, $\mathbf{E} \times \mathbf{H}$ away; H flips across, E doesn't.
- Field at (z,t) = source at $t - |z|/v$; front at $|z| = vt$.
- **vs** t: source shape, delayed. **vs** z: reversed, first part farthest out. Travel $-z$: $E(z,t) = E(0, t + z/v)$.
- Phase change $\Delta\phi$ in time Δt at one point: $\omega = \Delta\phi/\Delta t$, then $\lambda = 2\pi v/\omega$. Over a distance Δz at one time: $\beta = \Delta\phi/\Delta z$.
- Valid wave: depends on $t \mp z/v$ only, with $v = 1/\sqrt{\mu\epsilon}$; $\cos\omega t \cos\beta z$ is two waves (standing).

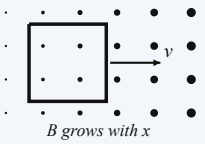
Ex 1 · Diagonal voltmeter



Square 2 m, $\dot{B}_z = +3$ T/s; $R_1 = R_2 = R_3 = 2\,\Omega$ (left, top, right). + lead at A.

- $\mathcal{E}_{ccw} = -(4)(3) = -12$ V; $I = -2$ A: 2 A clockwise (Lenz ✓).
- Diagonal: lower-right triangle holds half: $V + IR_3 = \mathcal{E}/2 \to V = -6 + 4 = -2$ V.
- Leads hugging R_3: $V = -IR_3 = +4$ V.

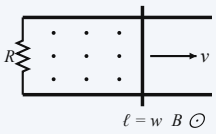
Ex 2 · Loop in a graded field



$s = 0.5$ m, $\mathbf{B} = 0.2(1 + x/2)\hat{z}$ T, $\mathbf{v} = 4\hat{x}$ m/s.

- $\Psi = s^2 B(x_c) \to \mathcal{E} = -s^2 B_0 v/L = -0.1$ V: clockwise (into stronger flux).
- Check: $vs\,\Delta B = 4(0.5)(0.05) = 0.1$ V ✓.

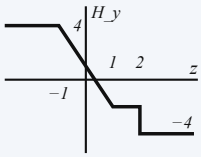
Ex 3 · Bar on rails



$B = 0.5$ T, $\ell = 0.4$ m, $v = 5$ m/s, $R = 2\,\Omega$.

- $\mathcal{E} = vB\ell = 1$ V, $I = 0.5$ A, $F = I\ell B = 0.1$ N against $\mathbf{v}$.
- $P = Fv = 0.5$ W $= I^2R$ ✓.

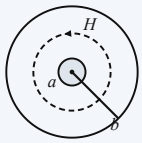
Ex 4 · Slab + sheet



$\mathbf{J} = 3\hat{x}\,[u(z+1) - u(z-1)] + 2\hat{x}\,\delta(z-2)$.

- Slab: $H_y = -3z$ in, ∓ 3 out. Sheet: $\frac{1}{2}(2\hat{x}) \times (\pm\hat{z}) = \mp\hat{y}$.
- $H_y(0) = 1$, $H_y(1.5) = -2$, $H_y(3) = -4$, $H_y(-2) = 4$ A/m. Net $K = 8$: far ± 4 ✓.

Ex 5 · Coax: L′, C′, v



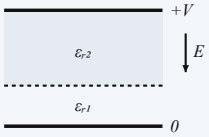
$b/a = 4$, filling $\epsilon_r = 2.25$.

- $L' = 2 \times 10^{-7} \ln 4 = 0.277\ \mu$H/m; $C' = \frac{2\pi\epsilon}{\ln 4} = 90.3$ pF/m.
- $L'C' = 2.5 \times 10^{-17} = \epsilon_r/c^2$ ✓, $v = 2 \times 10^8$ m/s.

Ex 6 · PP line and toroid; scaling

- $w = 5$ cm, $d = 1$ mm, $\ell = 2$ m: $L = \mu_0 d\ell/w = 50.3$ nH. $d \times 2$, $I \times \frac{1}{2}$: $B \times \frac{1}{2}$, $\Psi \times 1$, $L \times 2$.
- Toroid $N = 200$, $h = 2$ cm, $a = 1$, $b = 3$ cm: $L = 2 \times 10^{-7}(4 \times 10^4)(0.02)\ln 3 = 0.176$ mH; iron core $\mu_r = 100$: ×100.

Ex 7 · Two layers, then a leak



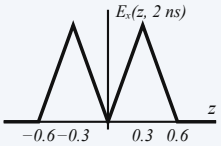
Air 1 mm under $\epsilon_r = 4$, 2 mm; 9 V.

- $D = \frac{9}{10^{-3}/\epsilon_0 + 2\times10^{-3}/4\epsilon_0} = 6000\epsilon_0 = 5.31 \times 10^{-8}$ C/m².
- $E_1 = 6000$, $E_2 = 1500$ V/m (6 + 3 = 9 V ✓); $P_2 = \frac{3}{4}D = 3.98 \times 10^{-8}$.
- Top layer leaks: $E_2 = 0$, $E_1 = 9000$ V/m, interface $\rho_s = 9000\epsilon_0 = 7.97 \times 10^{-8}$ C/m².

Ex 8 · Lossy coax; relaxation

- $G' = 5\ \mu$S/m, $\sigma = 10^{-5}$, $\epsilon_r = 4$: $C' = \epsilon G'/\sigma = 17.7$ pF/m, $\tau = \epsilon/\sigma = 3.54\ \mu$s. 10 V through $R_s = 1/G$: 5 V.
- $\rho(z,0) = \cos 2z$: $\rho = \cos(2z)e^{-t/\tau}$, $E_z = \frac{\sin 2z}{2\epsilon}e^{-t/\tau}$.

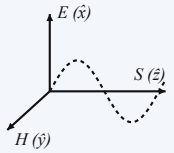
Ex 9 · Pulse from a sheet



$\mathbf{J}_s = -J(t)\hat{x}$, triangle 0 → 2 A/m at 1 ns → 0 at 2 ns; free space, $v = 0.3$ m/ns.

- $z = 0.45$ m, $t = 2$ ns: $\tau = 2 - 1.5 = 0.5$ ns, $J = 1$: $E_x = 60\pi = 188$ V/m, $H_y = +0.5$ A/m, $\mathbf{S}$ along $+\hat{z}$.
- Snapshot at 2 ns: pulse in $|z| < 0.6$ m, peak at $|z| = 0.3$ m.

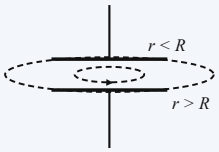
Ex 10 · Read a sinusoid



$\mathbf{E} = 10\cos(6\pi \times 10^8 t - 4\pi z)\hat{x}$.

- $f = 300$ MHz, $\lambda = 0.5$ m, $v = 1.5 \times 10^8$: $\epsilon_r = 4$, $\eta = 60\pi$.
- $\mathbf{H} = 0.0531\cos(\cdot)\hat{y}$ ($\hat{z} \times \hat{x} = \hat{y}$); $\langle S \rangle = \frac{100}{120\pi} = 0.265$ W/m².

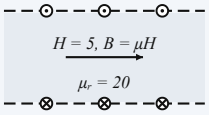
Ex 11 · H between capacitor plates



$R = 4$ cm, $d = 2$ mm, $V = 50\sin(2\pi \times 10^6 t)$.

- $\epsilon_0 \dot{E}_{\max} = \epsilon_0 \frac{V_0\omega}{d} = 1.39$ A/m².
- $H(2\text{ cm}) = \frac{r}{2}(1.39) = 0.0139$; $H(8\text{ cm}) = \frac{R^2}{2r}(1.39) = 0.0139$ A/m; $I_0 = 6.99$ mA.

Ex 12 · Magnetic slab; curl in cylindrical



- Sheets ± 5 A/m, gap filled with $\mu_r = 20$: $H = 5$ A/m (as in air), $B = 100\mu_0 = 1.26 \times 10^{-4}$ T, $M = 95$ A/m, bound sheets ± 95 A/m.
- Wire with $\mathbf{J} = J_0 \frac{r}{a}\hat{z}$: Ampère gives $H_\phi = \frac{J_0 r^2}{3a}$ inside. Check: $\frac{1}{r}\frac{d}{dr}\left(\frac{J_0 r^3}{3a}\right) = \frac{J_0 r}{a}$ ✓. Outside $H_\phi = \frac{J_0 a^2}{3r}$: curl 0 ✓.

Power

- $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ (W/m²), $S = E^2/\eta = \eta H^2$; $\langle S \rangle = \frac{E_0^2}{2\eta} = \frac{1}{2}\eta H_0^2$; $P = SA\cos\theta$; energy per area $\int S\,dt$.
- $\langle w \rangle = \frac{1}{2}\epsilon E_0^2$ ($w_e = w_m$), $S = wv$. A sheet supplies $\frac{\eta}{2}J_s^2$ per m² ($\frac{\eta}{4}J_s^2$ each way); $\mathbf{E} \cdot \mathbf{J}_s < 0$ there.

Traps

- cm² → m²: $\times 10^{-4}$. $\omega = 2\pi f$. ln, not log. $\mathcal{E} = -N\dot{\Psi}_1$, but $L \propto N^2$.
- $\tau = \epsilon/\sigma$ (Summer 2017 key: upside down). A big constant B gives no emf.
- Sheet: $\eta/2$ and $|z|$. Average power: the ½. H_t is continuous across a μ change, B_t is not.